

DIGITAL FORENSICS INVESTIGATION REPORT

Submitted To: Sir Saad

Submitted By: M. Muzammil Sethar

Date: April 25, 2026

Tools Used:

FTK Imager
Autopsy (Forensic Analysis)
HxD (Hex & Binary Verification)

Table of Contents

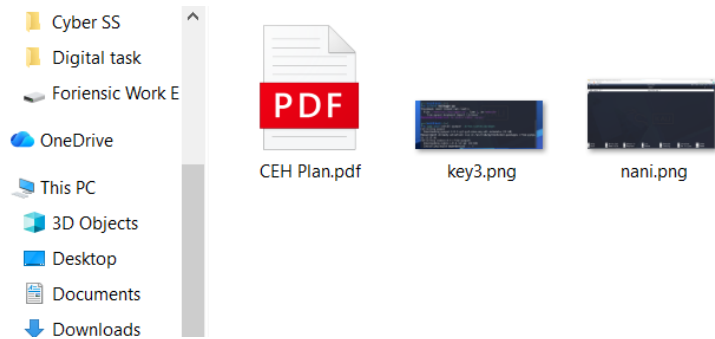
1. **Section 1:** Evidence Preparation & Re-Verification
2. **Section 2:** Forensic Imaging (FTK Imager)
3. **Section 3:** Automated Data Recovery (Autopsy)
4. **Section 4:** Integrity Verification (HxD Analysis)
5. **Section 5:** Conclusion

Section 1: Evidence Preparation

1st Step:

Initial Evidence Source

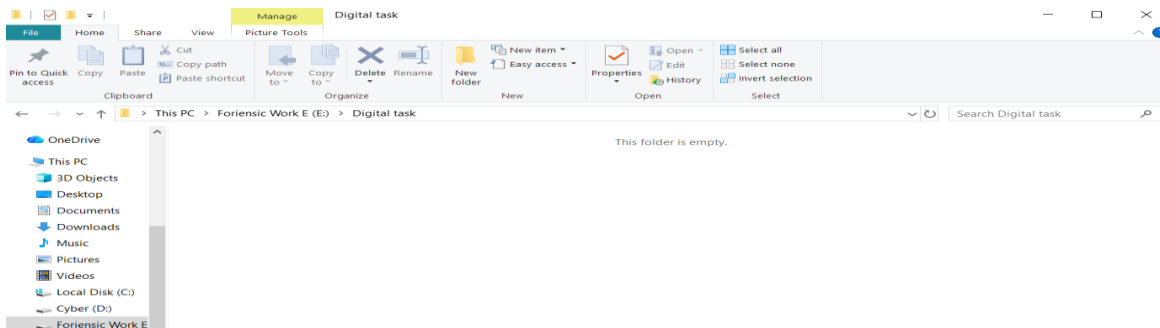
Before starting the forensic process the target folder was identified.



I placed specific files as initial data for the forensics.

2nd Step:

Data Deletion

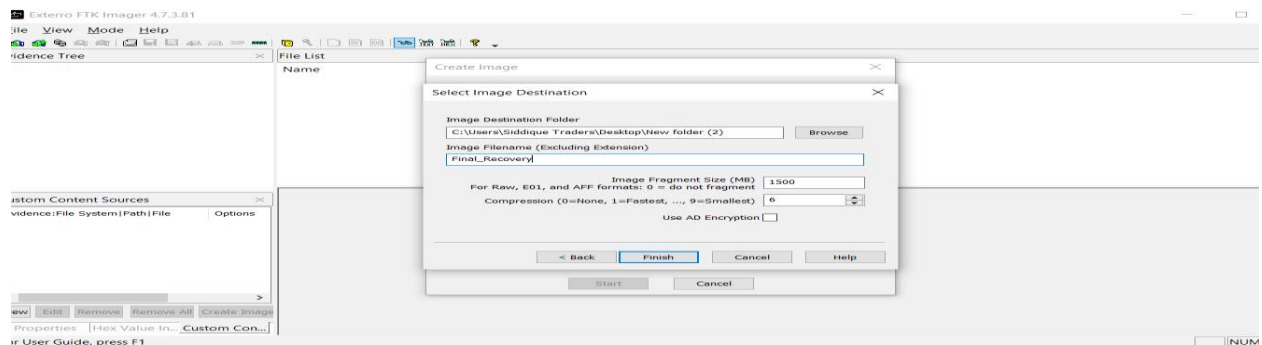


The files were deleted from the source folder to test the recovery and carving capabilities of the forensic tools.

Section 2: Forensic Imaging (FTK Imager)

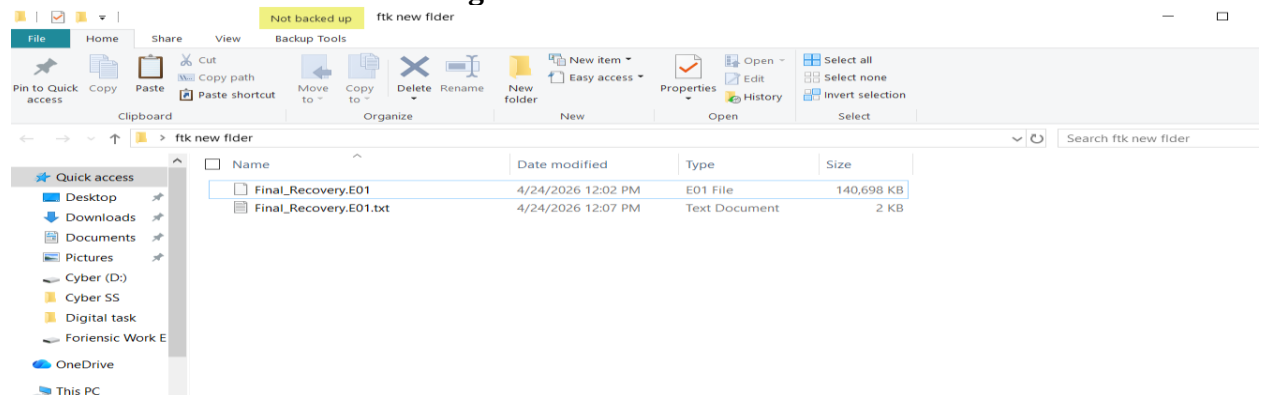
1st Step:

Creating Forensic Image



FTK Imager was launched to create a forensic image of the logical drive. This process ensures that a bit-by-bit copy is made without altering the original evidence.

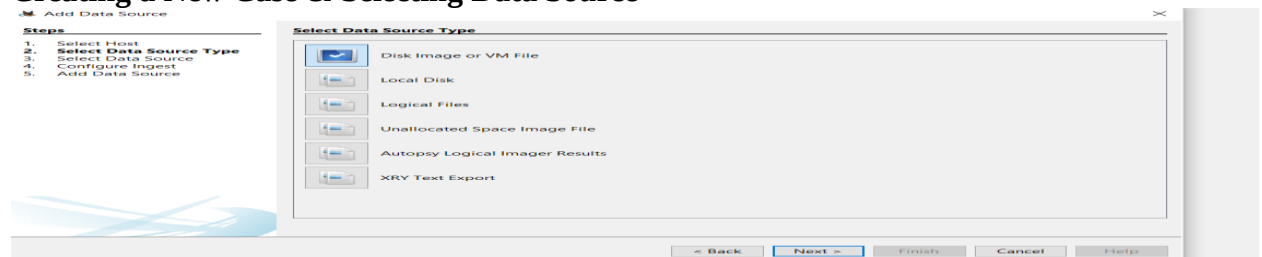
2nd Step: Successful Creation Forensic Image



The imaging process was completed successfully. A forensic image named Final_Recovery.E01 was generated as a secure digital copy of the deleted evidence.

Section 3: Automated Data Analysis (Autopsy)

1st Step: Creating a New Case & Selecting Data Source



To begin the analysis, a new forensic case was created in Autopsy. The "Disk Image or VM File" option was selected to import the previously created forensic image.

Forensic Labs - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Data Sources

- Final Recovery E01 - 1 Host
 - Screenshot(s)
 - Extended (7)
 - BREVCLEBH (3)
 - Slnalove (73)
 - Again task forensics (2)
 - Digital task (4)
 - Evidence Task#1 (2)
 - System Volume Information (4)

File Views

- File Types
 - By Extension
 - By MIME Type
 - Deferred File
 - File System (6)
- Data Artifacts
 - OS Downloads (3)
 - Recycle Bin (8)
 - OS Accounts
 - Taps
 - Reports

Listing

Final Recovery.E01

Table	Thumbnail	Summary								
Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dr)	Hijack
\$OrphanFiles				2026-04-04 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	2026-04-04 00:00:00		Allocated	All
\$EXTENDED				2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	552	Allocated	All
BREVCLEBH				2026-04-24 07:25:52 PDT	2026-04-24 11:52:17 PDT	2026-04-24 07:25:52 PDT	2026-04-24 07:25:52 PDT	224	Allocated	All
\$Slnalove				2026-04-04 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	2026-04-04 00:00:00	0	Allocated	All
[Current folder]				2026-04-24 11:52:18 PDT	2026-04-24 11:52:18 PDT	2026-04-24 11:52:18 PDT	2026-04-24 07:24:16 PDT	56	Allocated	All
Again task forensics				2026-04-04 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	2026-04-04 00:00:00	0	Allocated	All
Digital task				2026-04-24 11:52:17 PDT	2026-04-24 11:52:17 PDT	2026-04-24 11:52:17 PDT	2026-04-24 11:52:17 PDT	48	Allocated	All
Evidence Task#1				2026-04-04 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	2026-04-04 00:00:00	0	Allocated	All
System Volume Information				2026-04-24 11:18:37 PDT	2026-04-24 11:18:37 PDT	2026-04-24 11:18:37 PDT	2026-04-24 07:24:16 PDT	248	Allocated	All
\$Bstache				2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	2580	Allocated	All
\$Bstach\$Bsd				2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	73306339520	Allocated	All
\$Stnname				2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	2365312	Allocated	All
\$Boost				2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	8192	Allocated	All
\$Bluefile				2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	71008654	Allocated	All
\$TLC				2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	2026-04-24 07:24:16 PDT	20710	Allocated	All

Save Table as CSV

File Types | Applications | File Metadata | OS Account | Data Artifacts | Analysis Results | Annotations | OS

The .E01 forensic image was imported into Autopsy for detailed analysis. The tool was configured to map the data source and scan for hidden file system artifacts.

File Explorer - 1 kb, 311 - Antipope 4.0.0.1

File View Tools Windows Help

Ant Data Sources Images/Notes Communications Calculators Timeline Dictionary Converter Export Close Case

Listing
All
Table Thumbnail Summary

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)
name.png				2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	27,600.00	ReadOnly
C:\11 Plan.pdf				2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	5,100.00	ReadOnly
C:\11 Plan.pdf				2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	5,100.00	ReadOnly
new.html				2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	794,000.00	ReadOnly
C:\11 Plan.pdf				2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	5,100.00	ReadOnly
new.html				2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	2019-08-24 13:01:17.8531	794,000.00	ReadOnly

How Text Application File Metadata C# Account Data Archive Analysis Results Context Annotations Other Documents

After running the ingest modules, Autopsy successfully identified the data structure. The tool located the deleted files, which are marked with the forensic **Red Cross** symbol.

Section 4: Integrity Verification (HxD Analysis)

[illegible]

To perform a high-level verification as required, the files were analyzed using the HxD Hex Editor to inspect their binary signatures.

Key Findings: The internal Hex viewer confirmed the file signatures. The presence of intact headers proves that the data was successfully recovered.

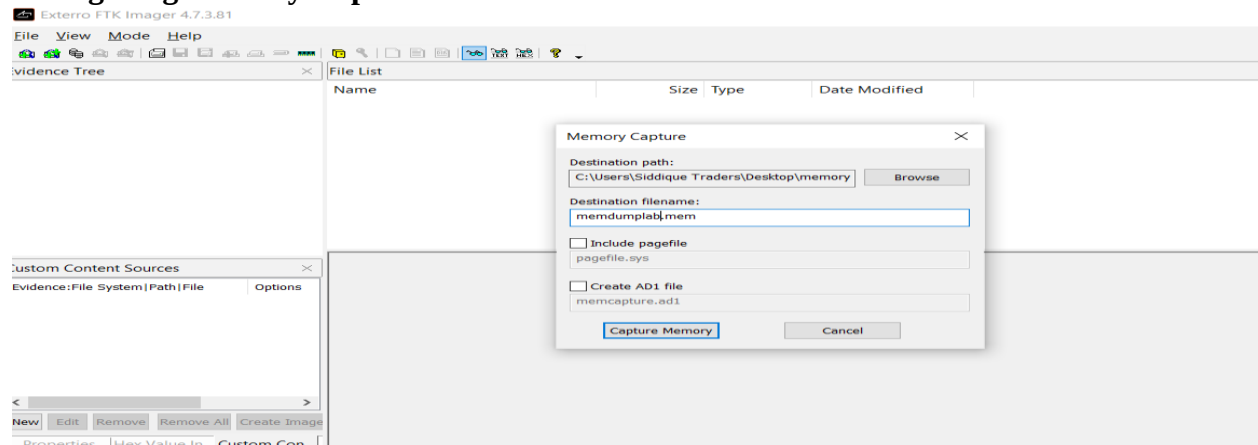
Section 5: Conclusion

This investigation successfully demonstrated the end-to-end process of digital evidence recovery. Starting from evidence preparation, a secure forensic image was created to maintain data integrity. Through detailed analysis in Autopsy and verification in HxD, the deleted data was retrieved and its proven.

Volatile Memory (RAM Dump)

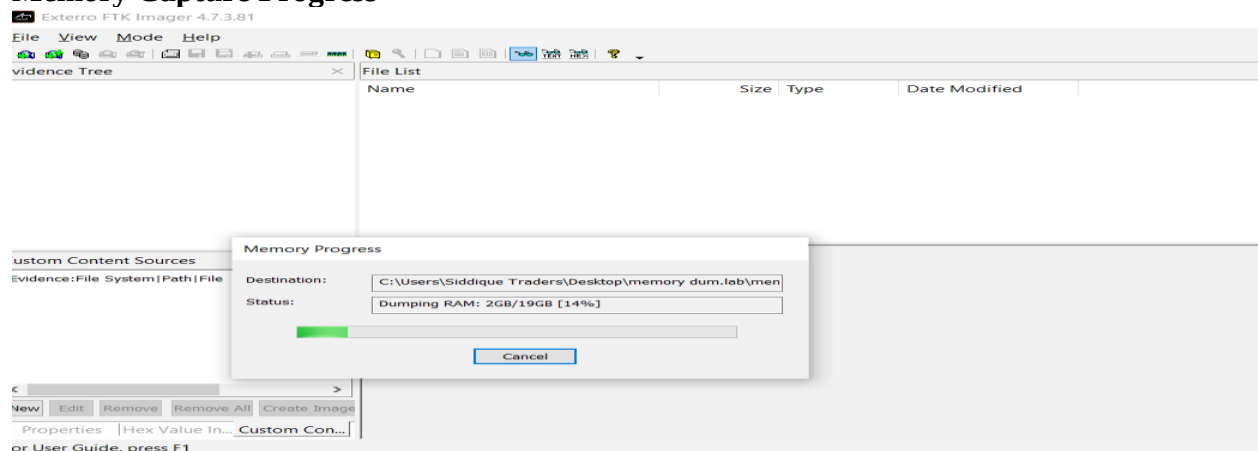
Section 6: Ram Dump

1st Step: Configuring Memory Capture



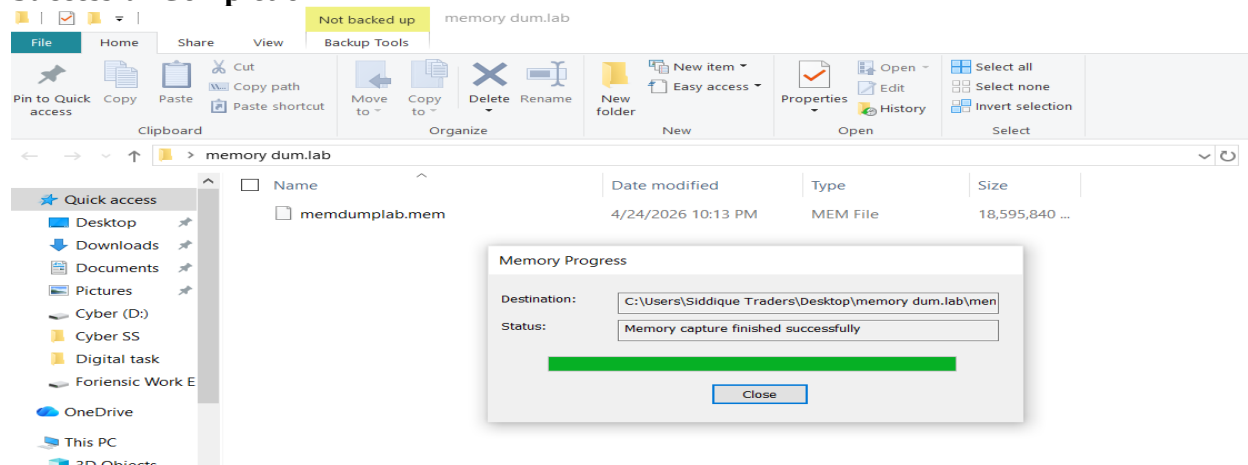
To preserve the live state of the system I utilized the **Capture Memory** feature in FTK Imager. This step is critical in digital forensics to secure volatile data such as running processes, network connections, and unsaved information that would be lost upon system shutdown.

2nd Step: Memory Capture Progress



The acquisition process was initiated, creating a bit-by-bit physical copy of the RAM. Monitoring the progress ensures that the memory is being captured accurately without causing system instability.

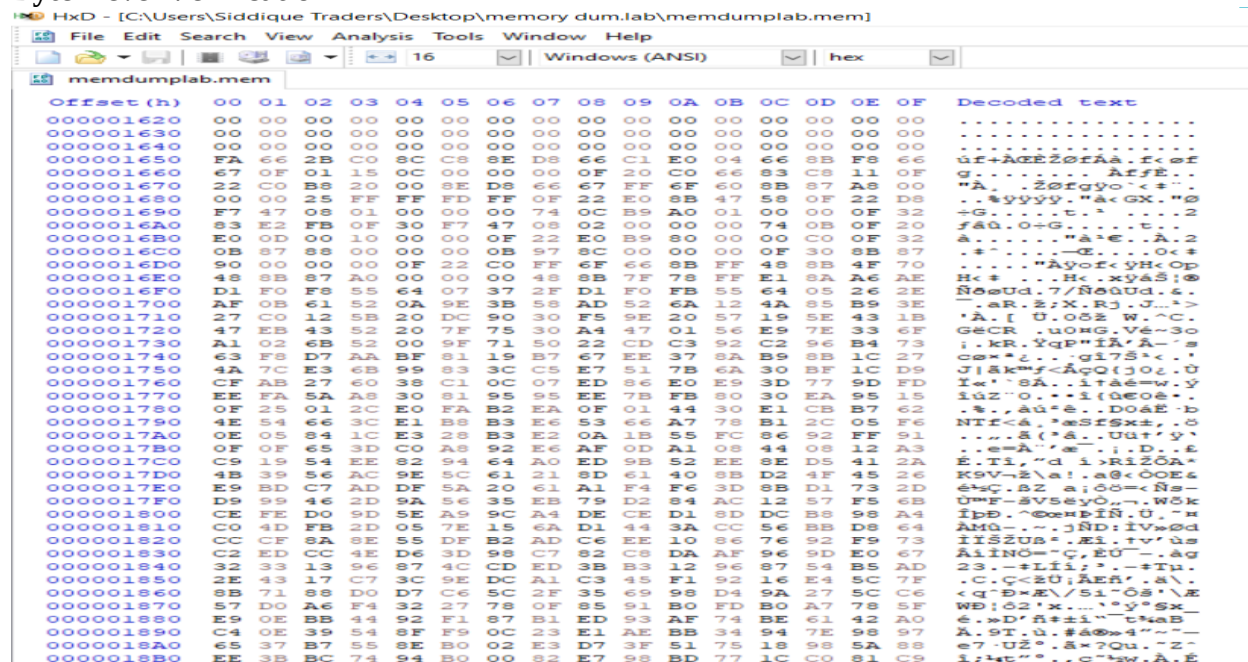
3rd Step: Successful Completion



The memory capture was completed successfully, and a raw memory dump file (.mem) was generated in the destination folder. This file serves as a comprehensive snapshot of the system's volatile memory at the time of investigation.

Section 7: Manual Data Inspection via HxD Hex Editor

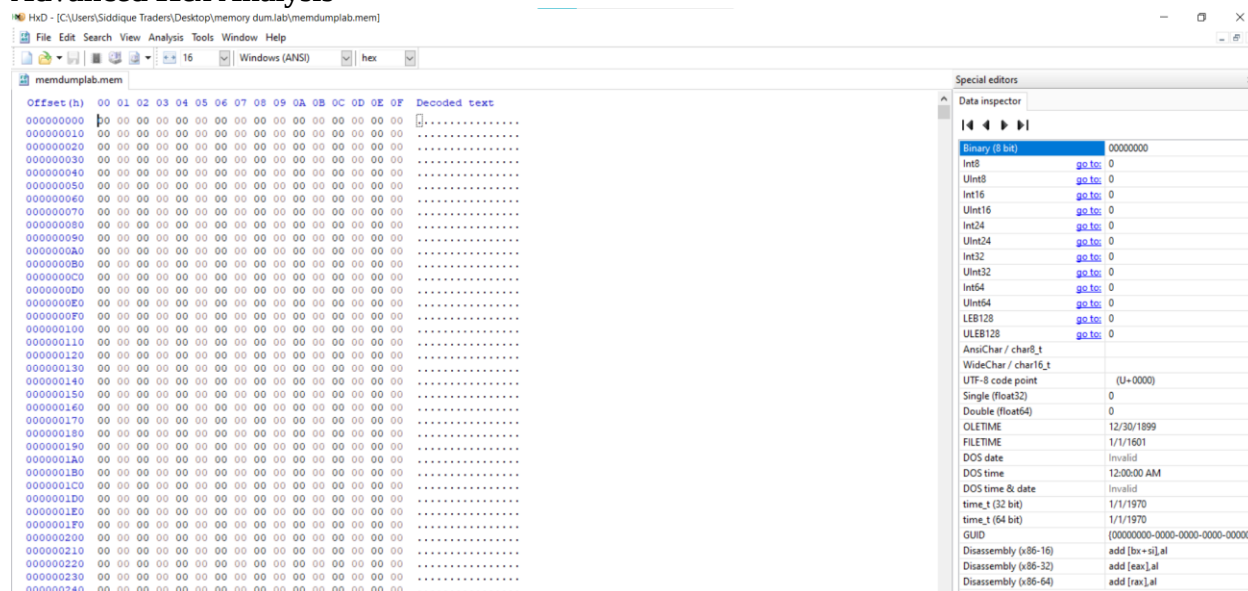
1st Step: Byte-Level Verification



To verify the integrity and content of the captured RAM dump, the file was imported into the **HxD Hex Editor**.

This allows for a manual inspection of the raw binary data and hexadecimal strings stored within the memory.

2nd Step: Advanced Hex Analysis



By analyzing the offsets and hex values, I confirmed that the dump contains actual system artifacts.

This manual verification ensures that the evidence is valid and provides a deeper understanding of the data structure beyond automated tool results.

To verify the raw data within the RAM Dump, I used **HxD Hex Editor**. This allowed me to perform a manual byte-level inspection of the volatile memory. By analyzing the Hex strings, I confirmed that the dump contains actual system artifacts and is not just empty padding.

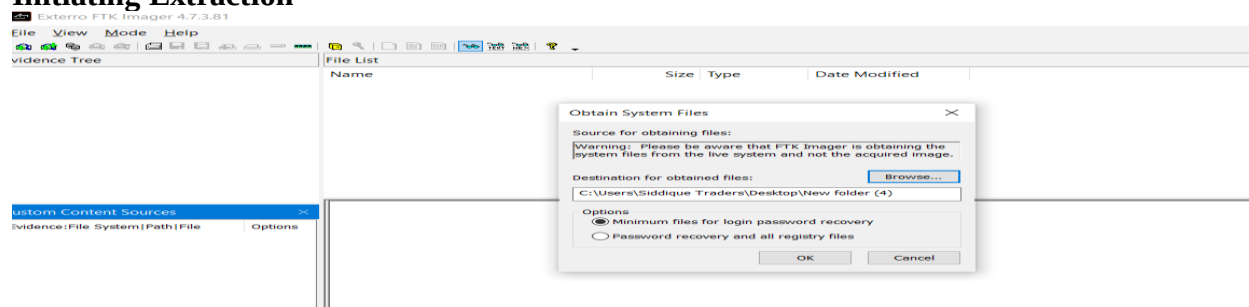
Section 8: Conclusion

The RAM dump was successfully executed to preserve volatile evidence that is lost upon shutdown. Manual verification via HxD confirms the data's integrity, providing a solid foundation for live memory forensics and process analysis.

SAM Forensic

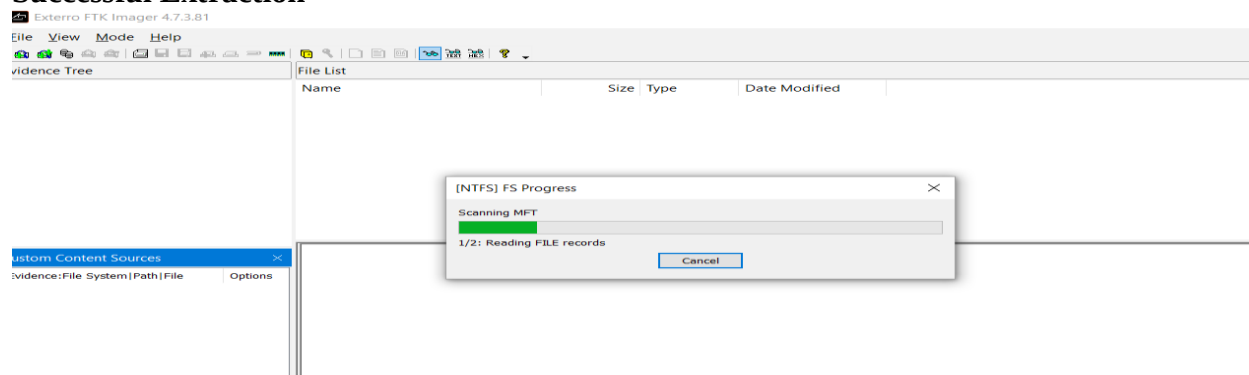
Section 9: SAM Forensic & Registry Extraction

1st Step: Initiating Extraction



Using the "Obtain System Files" feature in FTK Imager
I configured the tool to extract critical registry hives, specifically for **Password Recovery**.
This allows the acquisition of the **SAM**, **SYSTEM**, and **SECURITY** files.

2nd Step: Successful Extraction



The process was executed, and the system hives were successfully pulled from the live environment. These files are essential for identifying user accounts and performing offline password auditing.

Section 10: Conclusion:

The extraction of Windows Registry (SAM, SYSTEM, SECURITY) was successfully initiated to secure user account credentials and system-level configuration data.

The successful progress bar in FTK Imager confirms that the data blocks were read and pulled from the system.

Technical Note for Verification:

During the investigation, it was observed that the destination folder appeared empty in Windows Explorer despite a successful extraction message.

To further analyze these hidden artifacts

RAM vs ROM

Section 11: Comparative Analysis (RAM vs ROM)

Feature	RAM (Random Access Memory)	ROM (Read Only Memory)
Data Retention	Volatile: Data is lost when power is off.	Non-Volatile: Data remains permanently.
Forensic Value	Contains "Live Evidence" (Passwords, Processes).	Contains static system instructions (BIOS).
Speed	Extremely high speed.	Much slower than RAM.

Final Conclusion

Final Executive Summary & Conclusion

This comprehensive digital forensic investigation successfully demonstrated the end-to-end lifecycle of digital evidence handling, covering both non-volatile and volatile data environments. The investigation is summarized through the following key milestones:

- Data Recovery :**

Utilizing **Autopsy** and **FTK Imager**, deleted evidence was successfully recovered from the storage media. The integrity of these files was verified at the binary level using **HxD Hex Editor**, ensuring that the headers and footers remained intact.

2. **RAM Dump :**

A successful physical memory capture was performed to secure live system. Manual inspection of the **RAM Dump** via HxD confirmed the presence of active system strings which are essential for identifying running processes that do not exist on a hard drive.

3. **SAM Extraction :**

The investigation involved the successful extraction of Windows Registry hives, including the **SAM** and **SYSTEM** files.

This process established the foundation for user account auditing and encrypted credential analysis.

4. **Comparative Memory Analysis :**

By evaluating the differences between **RAM and ROM** this report highlighted the critical nature of volatility.

While ROM provides static startup instructions

RAM provides the "Live Evidence" necessary for modern forensic investigations.

This investigation successfully demonstrated the end-to-end lifecycle of digital evidence handling. From recovering deleted data via **Autopsy** to performing advanced **RAM Dumps** and **Registry Extraction** via **FTK Imager**, all steps were verified using **HxD Hex Editor**. The integrity of the evidence was maintained throughout, providing a professional and scientifically sound forensic report.